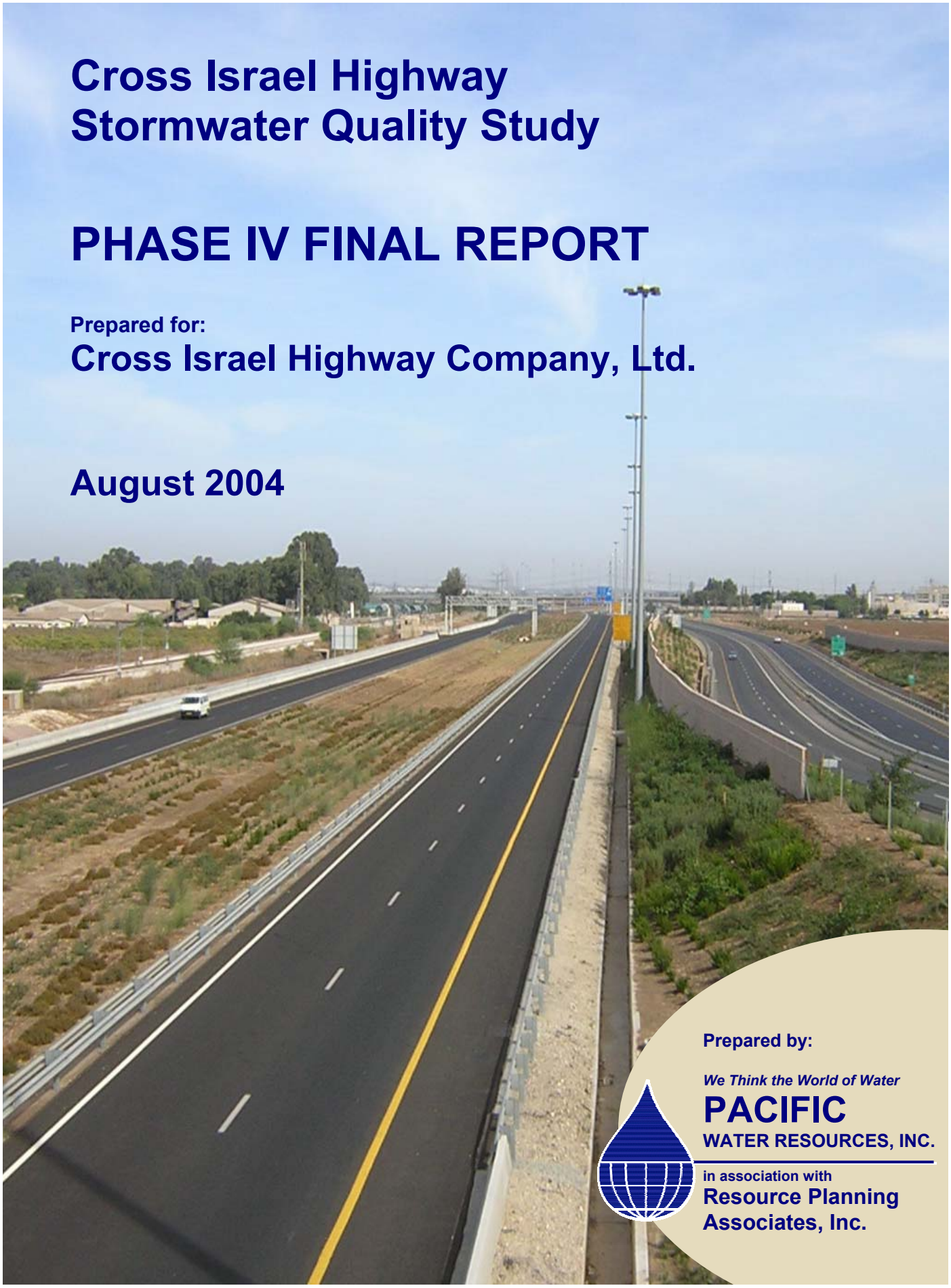
Cross Israel Highway Stormwater Quality Study

PHASE IV FINAL REPORT

Prepared for:

Cross Israel Highway Company, Ltd.

August 2004



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We Think the World of Water

PACIFIC
WATER RESOURCES, INC.

in association with
Resource Planning Associates, Inc.

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Cross Israel Highway Stormwater Quality Study

FINAL REPORT

EXECUTIVE SUMMARY

This document presents the final results of an extensive study of the stormwater quality from a 2.4 kilometer segment of Section 13 of the Cross Israel Highway (CIH), called for the purposes of this report, the Rosh Ha'ayin segment. In the spring of 2000, several parties raised a concern that runoff from this section of the CIH will impact the groundwater quality of the Hayarkon Springs. They petitioned the Supreme Court of Israel to issue an interim injunction against the continuation of the project until the issue of water contamination is resolved. The petitioners contended that runoff from the CIH, including toxic material spilled from traffic accidents may pollute the underground water resources of the Hayarkon Springs.

The Rosh Ha'ayin segment drains to the Einat and Rosh Ha'ayin Channels. These two channels flow northwesterly and westerly respectively from the vicinity of the CIH. The Rosh Ha'ayin Channel merges with the Einat Channel immediately adjacent to and north of Highway 483. The Einat Channel then passes adjacent to the well fields of the Hayarkon Springs after which it flows into the Hyarkon River.

A group of decision-makers, called the Accompanying Team (AT), was appointed by the National Planning and Building Board. The AT was to oversee the environmental aspects of the CIH, and was designated to deal with the problem. The responsibility of the AT was to develop a plan of action for the CIH to address these concerns and to oversee the implementation of the plan. The Team has developed a plan of action for the Rosh Ha'ayin segment referred to as the "Integrated Solution." Components relative to this study include the following:

- Design, construction and operation of spill containment facilities that will capture and contain any toxic spills that could occur along the 2.4 km of the CIH that drains to the vicinity of the Hayarkon Springs.
- Design and implementation of a monitoring plan that obtains the data needed to eventually address the various concerns of the Accompanying Team.
- Conduct a stormwater modeling study that: characterizes the stormwater runoff from the Rosh Ha'ayin segment; determines the specific pollutant reductions associated with roadway cleanings; and provides a recommendation on the type and cleaning frequency of the most appropriate highway cleaning practice, if needed.

The Supreme Court ruled in November 2001 that the construction of the CIH will not be halted and the "Integrated Solution" will be implemented. The required spill containment facilities were constructed before the CIH opened to traffic in the fall of 2002. Each of the ten containment facilities have a control gate that remains closed at all times to contain potential spills. The facilities are designed to contain a total spill from the largest tanker truck known to transport hazardous liquids in Israel. The containment gates are controlled from an operations center located 4 km south of the Rosh Ha'ayin segment. This center also operates a highway patrol for inspection and supervision.

PHASED APPROACH FOR STORMWATER QUALITY STUDY

In November 2001, Pacific Water Resources, Inc. in association with Resource Planning Associates was selected by Cross Israel Highway Company, Ltd. (CIHC) to conduct a study of the stormwater quality from the Rosh Ha'ayin segment and address the concerns of the Accompanying Team. Mr. Roger Sutherland, of Pacific Water Resources, Inc., is a recognized international expert in simulation modeling

of stormwater quality and sweeping practices. He is the author of the SIMPTM simulation model, used in this study. This model has several unique features, as explained in this report, but in particular is the only model that explicitly simulates the effectiveness of pavement sweeping or cleaning. Dr. Gary Minton of Resource Planning Associates has over 20 years of experience in the monitoring of stormwater and is a leading expert in stormwater treatment. He is the author of the first engineering textbook on stormwater treatment.

A phased approach was used. Phase I involved the collection and review of basic information regarding the design of the highway and the rainfall characteristics near the Rosh Ha'ayin segment. Phase I ended with the development of the Monitoring Plan dated October 2002.

Phase II involved the implementation of the monitoring plan and the interpretation of data gathered between October 2002 and July 2003 at locations along the Rosh Ha'ayin segment. The data was collected and presented by a team from the Nature Reserve Authority that was contracted by CIHC and supervised by the authors of this report. The monitoring program and the interpretations of the collected data were presented in a Monitoring Report, whose final draft was delivered to CIHC in May 2004.

The types of data collected during Phase II were the following:

- Quality of stormwater from the CIH pavements and containment basins
- Quality of stormwater in the Rosh Ha'ayin and Einat Channels
- Accumulation rates of road dirt on the CIH
- Accumulation rates of road dirt on two existing highways with high traffic volumes
- Physical and chemical characteristics of the road dirt on the three highways
- Performance of the Netivey Hamifratz pavement cleaner that were used to clean the highway, once it was open for traffic.

Eleven storms were monitored during the wet season of 2002-2003. Approximately 200 individual samples were taken and analyzed. Full detail of the monitoring procedures and results are found in the Monitoring Report (2004).

Phase III involved the use of a stormwater quality computer model called SIMPTM to simulate the pollutant accumulation and washoff by rainfall. The model was calibrated to best reproduce the stormwater and road direct accumulation data collected in Phase II. The calibrated model was used to simulate the pollutant conditions under two traffic volumes: 10,000 vehicles per day, the approximate volume when the data were collected in Phase II, and the future condition of 100,000 vehicles per day. SIMPTM was also used to evaluate the reduction benefits associated with various types of roadway cleaning practices.

Phase IV is the final documentation of the project Phase III modeling efforts and a recommended management program if necessary, which are the major topics of this report. The overall study goals were to determine if cleaning action is appropriate and to develop a management plan to mitigate, if needed, the potential water quality impacts associated with stormwater from the Rosh Ha'ayin segment.

POLLUTANTS OF INTEREST

Inherent to water quality planning is the establishment of a list of the pollutants of interest. Traditionally this is accomplished by the use of water quality standards. Water quality standards are generally specified as the maximum concentration a given pollutant can reach before a beneficial use is assumed to be impaired. Beneficial uses include human consumption, agricultural irrigation, industrial water supply, recreation like swimming, and aquatic species protection. Human consumption is regulated by drinking water standards. Other beneficial uses are commonly regulated by receiving water standards.

The stormwater samples collected during Phase II were analyzed for 35 analytes, identified by the Accompanying Team as possible pollutants of interest. The examination of data, presented in the Phase II report (May 2004), found that the concentrations of pollutants in the stormwater from the CIH are typical of that observed of highways in other countries for the current traffic conditions.

For the purposes of modeling using SIMPTM, the number of potential pollutants was reduced to seven. This reduction was done for two reasons. The first reason was to focus on those analytes that appear, based on the Phase II results, to exceed either the Israel Drinking Water Standards or the draft Kishon River Standards[1]. The second reason was to make the modeling task manageable. The selected analytes were:

- Total Suspended Solids (TSS)
- Total Chromium (TCr)
- Total Copper (TCu)
- Total Lead (TPb)
- Total Zinc (TZn)
- Total Nitrogen (TN)
- Total Phosphorus (TP)

TSS was selected because it is the primary pollutant simulated by SIMPTM. Other pollutants are related to this simulation. The four metals in the above list were selected based on the review of the stormwater data collected during Phase II. This review suggested that exceedances of either the drinking water or draft receiving water standards may occur at a traffic volume of 100,000 vehicles per day. However, the monitoring data indicated that at the low traffic volume during the monitoring period, of about 10,000 vehicles, only the chromium standard was exceeded. Both the chromium and lead concentrations exceeded the draft Kishon River Standards. Nitrogen and phosphorus were included as they are nutrients.

Considering that the concern regarding the CIH pavement discharges is related to the potential contamination of the drinking water supply of the Hayarkon Springs, it is most appropriate that the Accompanying Team focus on the Israeli Drinking Water Standards. However, consideration of the draft Kishon River receiving water standards was also included in this report.

MODELING HIGHWAY STORMWATER POLLUTION

Modeling provides an opportunity to simulate the site-specific pollutant accumulation and washoff processes that occur over time, including the removal of accumulated pollutants by various cleaning practices. By using a modeling approach, one can collect some data over a relatively short period of time on the important pollutant accumulation and washoff processes that occur. This information is used to calibrate the model, which is then used to simulate these accumulation and washoff behaviors over a much longer time frame of say several decades. But more importantly, a model can simulate conditions that do not currently exist, like for example, the pollutant accumulations and washoffs expected from the CIH once its ultimate traffic volume is reached.

The goals for the modeling of stormwater quality from the Rosh Ha'ayin segment were to determine (1) whether water quality standards are being exceeded once the highway's ultimate traffic is realized, (2) how frequently these exceedances will occur, and (3) how various cleaning practices implemented over time on the highway can reduce the frequency of the exceedances to some specified level.

1. The draft standards were prepared by the Ministry of Environment but have not been approved.

MODEL CALIBRATION AND SIMULATION TASKS

As stated previously, the model SIMPTM was used to simulate pollutant loadings and concentrations from both low and high traffic conditions. The key tasks in preparing and using the model were:

1. Obtain and analyze the historic rainfall record from a nearby rainfall gage to establish a simulation time period.
2. Analyze the dry weather road dirt data collected during the summer of 2003 to obtain the accumulation curves for both low and high traffic volumes on both traditional and porous asphalt pavements. Low traffic volumes were those that occurred on the CIH during the May-July 2003 monitoring period. High traffic volumes were those that occurred on Highways 1 and 4 during the same period.
3. Analyze the road dirt data to establish the particle size distributions and chemical qualities to be used in the model calibration in Task 4.
4. Using the accumulation curves for low traffic volumes from Task 2 and recent rainfall data from a nearby gage, calibrate the model to best represent the CIH wet weather road dirt accumulation and pollutant concentration data observed on both traditional and porous asphalt pavements from October 2002 through April 2003, the wet weather monitoring period.
5. Using the historic rainfall record from Task 1 and the calibrated model from Task 4, simulate the historic long-term storm-by-storm runoff volumes, pollutant concentrations and mass loadings for low traffic volumes.
6. Change the accumulation curves to those that reflect high traffic volumes from Task 2 and re-run the calibrated model from Task 4.
7. Analyze the simulated populations of pollutant concentrations and mass loadings from both Task 5 and Task 6 to obtain summary statistics and identify the frequency of water quality standard exceedances.
8. Use the model from Task 7 to evaluate the pollutant reduction benefits associated with various types and frequencies of street cleaning.
9. Use the model to estimate the pollutant mass load reductions associated with the alternative cleaning practices on both traditional and porous asphalt pavements under both low and high volume traffic conditions.

For the CIH Stormwater Quality Study, the long term hourly rainfall record from a station located at Kibbutz Einat immediately east of the Rosh Ha'ayin segment was used. This station provided a 57-year period of rainfall from January 1, 1938 through December 31, 1994 for the model simulations. A 57 year record of rainfall was used for the following reason. For any one future year its particular sequence of storms and storm depths is unknown. Using the historic rainfall record, it is assumed that any future year will mimic a previous historic year of record: that is the future will occur as the past, but the particular year in which this occurs is random and unknown. By simulating loadings based on the historic rainfall record the potential variation in loadings and concentrations is defined to the extent possible. The validity of this approach is recognized when considering that the rainfall during the monitoring period of October 2002 through April 2003 was abnormally and unusually wet. The rainfall was approximately 126% of normal. If only this one time period had been used for simulation, the loadings and concentrations would have been overstated.

MODEL RESULTS

Pollutant loadings are the mass of a given pollutant that is transported by stormwater from a designated area over a specified time period. They are in units of kilogram per curb-kilometer for highways. Pollutant loadings are very important since they can be used to compare the magnitude of pollution originating from a source of interest like the CIH to that of other source areas in the watershed such as agricultural lands, urban lands and other highways. Reductions in pollutant loadings can represent the water quality benefits of alternative cleaning practices using the SIMPTM model. The corresponding reduction in the pollutant concentrations, also simulated with SIMPTM can be compared to water quality standards.

The SIMPTM model was used to simulate over the historic rainfall record the storm-by-storm pollutant mass loadings and concentrations. This was done for both current (i.e. low) and ultimate (i.e. high) traffic conditions with no cleaning operations. Table ES-1 presents the average annual pollutant loadings and concentrations for pavement discharges from the Rosh Ha'ayin segment over the 57-year simulation period. The table shows that a ten-fold increase in traffic volume will, depending on the pollutant, result in a 291% to 315% increase in pollutant mass loadings discharged annually from the Rosh Ha'ayin segment of the CIH.

Table ES-1 Average Annual Pollutant Loadings[a] and Mean Concentrations for the CIH Under Low[b] and High[c] Traffic Volume Conditions with No Cleaning

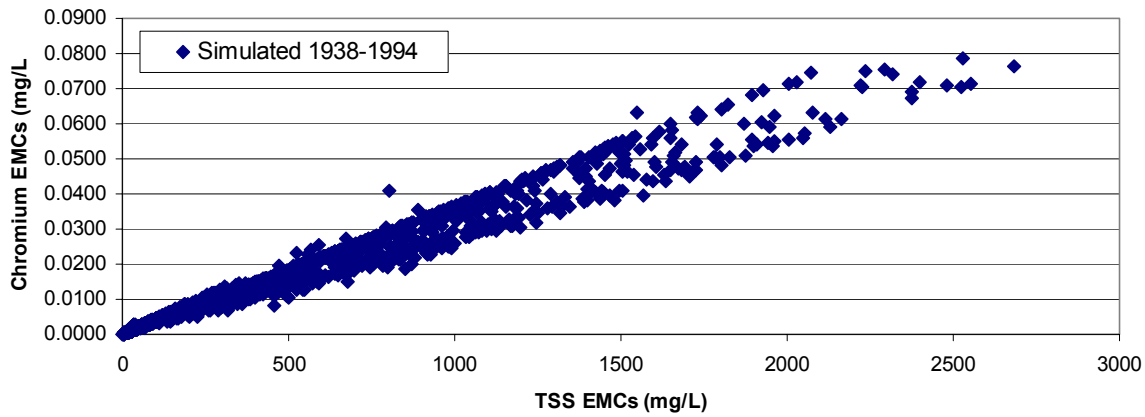
Pollutant	LOADINGS		Percent Increase	CONCENTRATIONS		Drinking Standards Mg/L	Draft River Standards Mg/L
	Low Traffic	High Traffic		Low Traffic	High Traffic		
TSS	498	1976	297	112	443	NS[d]	NS
TCr	.017	.067	294	.021	.032	0.05	0.01
TCu	.012	.048	300	.018	.026	1.40	0.05
TPb	.056	.231	313	.013	.052	0.10	0.01
TZn	.054	.224	315	.083	.121	5.00	1.00
TN	1.63	6.37	291	.365	1.43	NS	NS
TP	.301	1.22	305	.067	.274	NS	NS

a kg/curb-km/yr
b Low Traffic is approximately 10,000 vehicles per day
c High Traffic is approximately 100,000 vehicles per day
d. NS means no standard

Table ES-1 indicates that the annual pollutant loadings and mean concentrations increase by only a factor of approximately three although the volume of traffic increases by a factor of 10. How is this possible? There are two reasons. First, the amount of road dirt that accumulates on highways is not directly proportional to the traffic volume. Secondly, at higher quantities of accumulated road dirt most of the storms have insufficient volumes to remove all of the accumulated material. This phenomenon is further discussed in the report, using data collected during the monitoring.

Presented in this Executive Summary are the findings for only chromium with respect to exceedances of water quality standards. Chromium is selected because the data collected in Phase II suggest that of the seven pollutants, it is likely to have the highest frequency of exceedance of either the Israel Drinking Water Standards or the draft Kishon River Standards. Figure ES-1 presents the paired TSS and chromium concentrations simulated for the 2037 storm discharges throughout the 57-year simulation period from traditional pavements under high traffic volume conditions and with no road cleaning occurring.

Figure ES-1 Simulated TSS and Chromium Event Mean Concentrations from Traditional CIH Pavements with High Traffic Volume



Model simulation indicates that chromium concentrations in pavement discharges from the Rosh Ha'ayin segment under the projected conditions of approximately 100,000 vehicles per day will exceed the Israeli drinking water standard of 0.05 mg/L for approximately 13% of the storm events, an average of five times each year. In other words, the Israeli drinking water standard for chromium will be complied to approximately 87% of the time at the projected traffic volume of 100,000 vehicles per day. This high compliance frequency is achieved without pavement cleaning. The model results also indicate that the concentration of chromium in stormwater directly from the highway pavements will likely exceed the draft Kishon River Standards during 100% of the storm events. Regarding lead, copper and zinc: the simulations indicate that the frequency of standard exceedances for lead are similar to chromium, on the order of 13%. However, this simulation does not recognize the long-term effects of the use of unleaded gas, which will diminish the concentrations of lead in stormwater. Hence, the simulations overstate the likely frequency of future exceedances. In contrast, the concentrations of copper and zinc did not exceed the drinking water standard in any of the 2037 simulated storms.

DISCUSSION

Two primary questions of key importance should be addressed**.**

Question 1: Will reducing the frequency of the water quality standards exceedances significantly improve the quality of water in the Einat and Rosh Ha'ayin Channels?

In the monitoring study, Phase II, stormwater samples were taken from the two channels at stations upstream of the Rosh Ha'ayin segment. The data indicate that with respect to chromium the Israeli Drinking Water Standards were exceeded in four of 16 samples, or 25%, and the draft Kishon Standards were exceeded in 13 of 16 samples, or 81%. As previously stated, the model simulations indicate the chromium standard may be exceeded 13% and 100% for the drinking water and draft river water standards respectively.

The Rosh Ha'ayin segment pavement represents only 1% of the total watershed area that drains to Einat and Rosh Ha'ayin Channels. Other uncontrolled sources of pollution exist in the watershed that are much greater in area and pollutant loading, such as agriculture, highways, and commercial and residential areas. Failure to control these other sources would negate any benefits associated with reducing the pollutants from the CIH.

Therefore, it is concluded that reducing the frequency of exceedances from the Rosh Ha'ayin segment will not likely improve water quality in the Einat and Rosh Ha'ayin Channels. Furthermore, in the absence of any national Israeli policy on the control of pollutants in stormwater from all nonpoint sources, it would be a waste of public money to treat stormwater from the Rosh Ha'ayin segment while not addressing the other more dominant discharges.

Question 2: If we wish to increase the compliance frequency for the water quality standards, what highway cleaning practices would be needed?

For the Israeli Drinking Water Standards with respect to chromium, three cleanings during each wet season with a regenerative air or a modern vacuum street sweeper would be needed to increase compliance from 87%, that already exists without sweeping, to 90%. Increasing compliance to 95% would require that the pavement be swept six times during each wet season with either type of machine.

For the draft Kishon River Standards, highway cleaning practices alone would not be able to increase the compliance frequency. This is because cleaning practices are only marginally effective at removing dissolved pollutants. The monitoring data indicate that the mean and median concentrations of dissolved chromium are 0.017 and 0.011 mg/L, respectively, under low traffic volume. These values exceed the proposed standard of 0.010 mg/L. To meet the standards at a reasonable frequency of compliance would therefore require both extensive sweeping and a stormwater treatment system that is capable of removing dissolved metals at considerable expense**.**

CONCLUSIONS

The review of the monitoring data and model simulation with SIMPTM leads to the following conclusions.

1. The quality of stormwater and the accumulation of road dirt and its chemistry from the Rosh Ha'ayin segment is typical of freeways in other countries with similar traffic volumes.

2. It is estimated that without pavement cleaning, the quality of the stormwater will over the long term comply with the Israeli Drinking Water Standards for 87% of all storms, but 0% with respect to the draft Kishon River standards.

3. Currently, the observed quality of water in the Einat and Rosh Ha'ayin channels above the CIH fails to meet either the drinking water or the draft river standards at similar frequencies to the stormwater from the CIH.

4. Compliance of the stormwater from the CIH with the drinking water standards can be increased from 87% to 90% by cleaning the Rosh Ha'ayin segment three times per year during the wet season; 95% if cleaned six times per wet season each year.

5. Because the Rosh Ha'ayin segment pavement represents only 1% of the watershed area draining to the two channels, cleaning the CIH alone will not provide noticeable water quality benefits to the flow in either channel.

6. The containment basins and other safety practices by CIHC and Derech Eretz, Ltd (the company that is responsible for operating the highway) provides satisfactory protection of the potential harmful effects of hazardous chemical spills from accidents on the CIH.

RECOMMENDATIONS

1. Absent a national or watershed policy that leads to the reduction of non-point pollution from all sources in the watershed of the Einat and Rosh Ha'ayin channels, it is recommended that the Rosh Ha'ayin segment of the CIH not be cleaned or swept at this time.

2. If it is decided to clean the Rosh Ha'ayin section of the CIH, the selected cleaning equipment, if other than the Netivey Hamifratz pavement cleaner, should be tested to verify its performance. In addition, hand cleaning of the gutters along the sections of porous pavement should be conducted in the same manner that occurred when the highway was cleaned during the monitoring period.

3. To verify the simulations of SIMPTM, it is recommended that the CIHC conduct further monitoring when the traffic volume reaches approximately 100,000 vehicles per day. The types of data collected should be specific to the needs of verifying the model simulations presented in this report. The detailed monitoring plan should be prepared when the volume of traffic approaches 100,000 vehicles.

4. Current measures to protect against the potential harmful effects of accidental hazardous chemical spills should be continued.

Cross Israel Highway Stormwater Quality Study

FINAL REPORT

August 2004

1.0 INTRODUCTION

Looking North from Highway 483 Overpass

This document presents the final results of an extensive study of the stormwater quality from a 2.4 kilometer segment of the Cross Israel Highway (CIH), called for the purposes of this report, the Rosh Ha'ayin segment. In the spring of 2000, several parties including environmental groups and biologists raised a concern that runoff from this section of the CIH will impact the groundwater quality of the Hayarkon Springs. They petitioned the Supreme Court of Israel to issue an interim injunction against the continuation of the project until the issue of water contamination is resolved. The petitioners contended that runoff from the CIH, including toxic material spilled from traffic accidents, may pollute the underground water resources of the Hayarkon Springs.

The Rosh Ha'ayin segment drains to the Einat and Rosh Ha'ayin Channels. These two channels flow northwesterly and westerly respectively from the vicinity of the CIH. The Rosh Ha'ayin Channel merges with the Einat Channel immediately adjacent to and north of Highway 483. The Einat Channel then passes adjacent to the well fields of the Hayarkon Springs after which it flows into the Hyarkon River.

The stormwater quality issues raised by these environmental groups have never been raised in Israel before so, in September 2000, Mr. Reuven Levon, Chief Engineer of Cross Israel Highway Company, Ltd. (CIHC) contacted Pacific Water Resources, Inc. (PWR) and requested a technical memorandum on the pollution characteristics of highway runoff and its mitigation. The November 2000 memorandum written by PWR is included as Appendix A.

Ultimately, a group of decision-makers, called the Accompanying Team (AT), was appointed by the National Planning and Building Board. The AT was to oversee the environmental aspects of the CIH, and was designated to deal with the problem. The responsibility of the AT was to develop a plan of action for the CIH to address these concerns and to oversee the implementation of the plan. The Team has developed a plan of action for the Rosh Ha'ayin segment referred to as the "Integrated Solution." Components relative to this study include the following:

- Design, construction and operation of spill containment facilities that will capture and contain any toxic spills that could occur along the Rosh Ha'ayin segment that drains to the vicinity of the Hayarkon Springs.

- Design and implementation of a monitoring plan that obtains the data needed to eventually address the various concerns of the Accompanying Team.
- Conduct a stormwater modeling study that: characterizes the stormwater runoff from the Rosh Ha'ayin segment; determines the specific pollutant reductions associated with roadway cleanings; and provides a recommendation on the type and cleaning frequency of the most appropriate highway cleaning practice, if needed.

The Supreme Court ruled in November 2001 that the construction of the CIH will not be halted and the "Integrated Solution" will be implemented immediately. Anticipating this decision, the Derech Eretz Construction Joint Venture (CJV) completed the design of the required spill containment facilities that were constructed before the CIH opened to traffic in the fall of 2002. The ten containment facilities each have a control gate that remains closed at all times to contain potential spills. The facilities are designed to contain a total spill from the largest tanker truck known to transport hazardous liquids in Israel along with the estimated runoff volume from a 2-year rainfall event. A 2-year rainfall event is an intense rainfall whose depth will occur once every two years on the average. Once a rainfall event has ended and the operations personnel confirm that no spill occurred, the gates are automatically opened (from a control center that monitors them continuously) and the capture runoff volume is released.

Dam and Control Gate for Containment G2-R along Northbound Lanes

These containments demonstrate CIHC's commitment to clean water and their understanding of what a devastating impact that such a rare occurrence could have on the nation's water supply. Building facilities designed to contain a total spill from a large tanker truck along with the runoff from a 2-year rainfall event is quite rare throughout the world.

PWR/RPA is aware of only one location in the United States where spill containment has been incorporated into the design of highways and other land developments to protect an underground drinking water supply. That location is the South Shore Columbia Well Field in Portland, Oregon USA where highways and other development was planned for lands located over groundwater recharge areas of Portland's emergency back-up water supply located on the south shore of the mighty Columbia River.

In November 2001, Pacific Water Resources, Inc. (PWR), in association with Planning Resources Associates (RPA), was selected by Cross Israel Highway Company, Ltd. (CIHC) to conduct a study of the stormwater quality from the Rosh Ha'ayin segment and address the concerns of the Accompanying Team.

Mr. Roger Sutherland, of Pacific Water Resources, Inc., is a recognized international expert in simulation modeling of stormwater quality and sweeping practices. He is the author of the SIMPTM simulation model, used in this study. This model has several unique features, as explained in this report, but in particular is the only model that explicitly simulates the effectiveness of pavement sweeping or cleaning. Dr. Gary Minton of Resource Planning Associates has over 20 years of experience in the monitoring of

stormwater and is a leading expert in stormwater treatment. He is the author of the first engineering textbook on stormwater treatment.

2.0 PROJECT GOALS, OBJECTIVES AND PHASES

The CIH section of interest is described as from the Kesem Interchange (Highway 5) to the Nahshonim Interchange (Highway 471). It is referred to as the Rosh Ha'ayin segment. The overall study goals were to determine if cleaning actions are needed and to develop a management plan to mitigate, if necessary, the potential water quality impacts associated with stormwater from the Rosh Ha'ayin segment of the CIH, and. To meet these goals the following objectives of the study were identified during the preparation of the Monitoring Plan:

1. Characterize the quality of the stormwater.
2. Evaluate the effect of pavement cleaning on the quality of the stormwater runoff.

A Phased approach was used by the study. Phase I involved the collection and review of basic information regarding the design of the highway and the rainfall characteristics near the Rosh Ha'ayin segment. The study's principal investigators, Mr. Roger Sutherland, PE, of PWR and Dr. Gary Minton, PhD, PE, of RPA, traveled to Israel in January 2002 to meet with CIHC, CJV and the Accompanying Team to discuss the stormwater study and the design of the monitoring plan. Phase I ended with the development of the Monitoring Plan dated October 2002.

Phase II involved the implementation of the monitoring plan and the interpretation of data gathered between October 2002 and July 2003 at locations along the Rosh Ha'ayin segment. The monitoring program and the interpretations of the collected data were presented in a Monitoring Report, whose first draft was produced in September 2003. Dr. Minton traveled to Israel in October 2003 to discuss the report with the Accompanying Team. The final version of the Monitoring Report was delivered to CIHC in May 2004.

Phase III involved the use of a stormwater quality simulation program call SIMPTM to simulate the pollutant accumulation and washoff by rainfall under the existing condition of low traffic volume that occurred during the summer of 2003 and the future conditions where the ultimate volume will increase to an estimated 100,000 vehicles per day. The model was calibrated to best reproduce the data collected in Phase II and to simulate the pollutant reduction benefits associated with various types of roadway cleaning practices.

Phase IV is the final documentation of Phase III modeling efforts and the recommended management program, which are the major topics of this report. This report addresses both of the study objectives listed previously. The first objective was initially addressed in the Phase II Monitoring Report. Another objective of the Phase II Monitoring Plan was to provide data to assist in the evaluation of the effects of pavement cleaning on the stormwater quality of the Rosh Ha'ayin segment. Key to the analysis is the SIMPTM model (Sutherland and Jelen, 1998). It was used to estimate storm-by-storm and thereby annual washoff loadings of key pollutants with and without pavement cleaning. Much of the data collected in Phase II was designed for the calibration of the model. The results of the model calibration and the modeling analysis are presented later in this report.

3.0 DESCRIPTION OF THE ROSH HA'AYIN SEGMENT OF THE CIH

As stated previously, the Rosh Ha'ayin segment of the CIH is close to Hayarkon Springs. It is located in the eastern fringe of the metropolitan area of Tel Aviv, as shown in Figure 1. It consists of two southbound and two northbound lanes, which are separated. Most of the Rosh Ha'ayin segment is on fill. New Jersey barriers are situated along the perimeter of the fill areas, the purpose of which is to keep sediments on the highway. Water drains from, or though in the case of porous pavements, to small (i.e. 30 cm wide) drainage collection channels or gutters located at the edge of the pavement adjacent to the New Jersey barriers. At approximately every 75 meters (m), the water is discharged down a paved portion of the highway slope to a channel that transports the water to a containment basin. As stated previously, the purpose of the containment basin is to retain hazardous spills along with stormwater runoff up to the estimated runoff volume for a 2-year or 50% probability event.

Looking North Along Southbound Porous Pavement Section near Pavement Sampler SP20

Drainage Channel Along Northbound Porous Pavement Section near G5-R

The Rosh Ha'ayin segment of the CIH has both traditional and porous pavements. The porous pavement is not fully porous. Rather, only the final lift of approximately 40 millimeters (mm) is porous. Its purpose is to reduce noise and wet weather skid potential. To reduce costs, the porous pavement is used only in the immediate vicinity of populated areas. Hence, the monitoring was structured to ascertain possible differences in road dirt accumulations, washoff loadings, speciation of pollutants, and effects of cleaning practices between the two types of pavement. During the period of monitoring, the northbound shoulders were periodically cleaned.

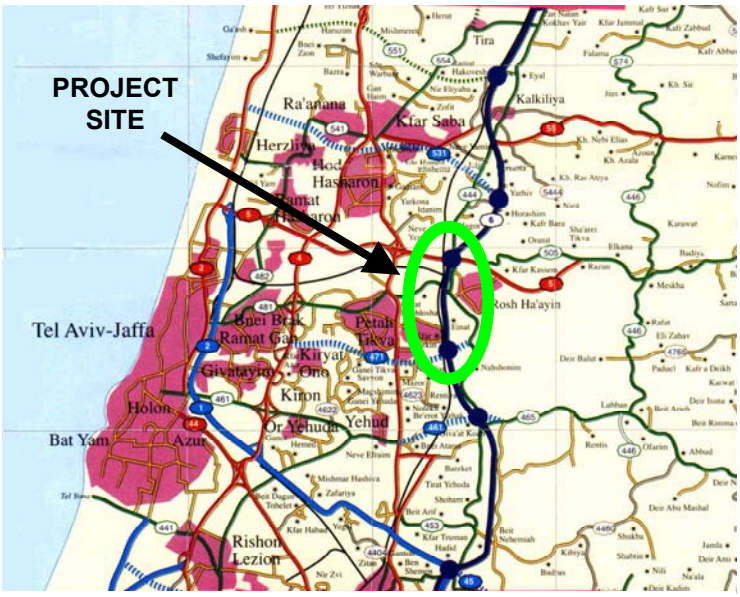


Figure 1 The Rosh Ha'ayin Segment Vicinity Map

4.0 KEY FINDINGS OF THE MONITORING STUDY

The types of data collected during Phase II were the following:

- Quality of stormwater from the CIH pavements and containment basins
- Quality of stormwater in the Rosh Ha'ayin and Einat Channels
- Accumulation rates of road dirt on the CIH
- Accumulation rates of road dirt on two existing highways with high traffic volumes
- Physical and chemical characteristics of the road dirt on the three highways
- Pick-up performance of the Netivey Hamifratz pavement cleaner

Eleven storms were monitored during the wet season of 2002-2003. Approximately 200 individual samples were taken and analyzed. Full detail of the monitoring procedures and results are found in the Monitoring Report (2004). As stated previously, Phase III involved the use of a stormwater quality computer model called SIMPTM to simulate the pollutant accumulation and washoff by rainfall. The model was calibrated to best reproduce the stormwater and road dirt accumulation data collected in Phase II. The calibrated model was used to simulate the pollutant conditions for two traffic volumes: 10,000 vehicles per day, the approximate volume when the data were collected in Phase II, and the future condition of 100,000 vehicles per day. SIMPTM was also used to evaluate the reduction benefits associated with various types of roadway cleaning practices.

The stormwater samples collected during Phase II were analyzed for 35 analytes, identified by the Accompanying Team as possible pollutants of interest. The examination of data, presented in the Phase II report (May 2004), found that the concentrations of pollutants in the stormwater from the CIH are typical of that observed of highways in other countries for the current traffic conditions.

5.0 POLLUTANTS OF INTEREST

Inherent to water quality planning is the establishment of a list of the pollutants of interest. Traditionally this is accomplished by the use of water quality standards. Water quality standards are generally specified as the maximum concentration a given pollutant can reach before a beneficial use is assumed to be impaired. Beneficial uses include human consumption, agricultural irrigation, industrial water supply, recreation like swimming, and aquatic species protection. Human consumption is regulated by drinking water standards. Other beneficial uses are commonly regulated by receiving water standards.

The AT identified the analytes in Table 1 as possible pollutants of interest (Phase II Monitoring Report, May, 2004). Of these, the monitoring of stormwater from the Rosh Ha'ayin segment found that 31 analytes exceeded the detection limit in at least 10% of the samples of stormwater. These 31 analytes are listed in Table 2 with the detection frequency for each (Phase II Monitoring Report, May 2004).

For the purposes of modeling the number of potential pollutants was further reduced to seven for two reasons. The first reason was to focus on those analytes that appear, based on the Phase II results, to exceed either the Israel Drinking Water Standards or the draft Kishon River Standards[1]. The second reason was to make the modeling task manageable.

[1] The draft standards were prepared by the Ministry of Environment but have not been approved.

Table 1 Stormwater Analytes Examined in this Study

Alkalinity	Dissolved zinc	Total phosphorus
Ammonia (NH4)	Hardness	Total recoverable cadmium
BETX[a]	Oil and Grease	Total recoverable chromium
Chemical oxygen demand (COD)	Nitrite (NO2)	Total recoverable cobalt
Conductivity	Nitrate (NO3)	Total recoverable copper
Dissolved cadmium	pH	Total recoverable iron
Dissolved chromium	Soluble reactive phosphorus	Total recoverable lead
Dissolved cobalt	Temperature	Total recoverable vanadium
Dissolved copper	Total Kjeldahl nitrogen (TKN)	Total recoverable zinc
Dissolved iron	Total organic carbons (TOC)	Total suspended solids (TSS)
Dissolved lead	Total petroleum hydrocarbons (TPH)	Turbidity
Dissolved vanadium		

a Benzene, Ethyl Benzene, Toluene, Xylene

Table 2 Detection Frequencies

ANALYTE	DET FREQ	ANALYTE	DET FREQ	ANALYTE	DET FREQ
Alkalinity	100%	Dissolved zinc	100%	Total phosphorus	91%
Ammonia (NH4)	47%	Hardness	100%	Total chromium[a]	100%
COD	100%	Oil & Grease	13%	Total cobalt[a]	38%
Conductivity	100%	Nitrite (NO2)	100%	Total copper[a]	98%
Dissolved chromium	96%	Nitrate (NO3)	97%	Total iron[a]	100%
Dissolved cobalt	13%	pH	100%	Total lead[a]	58%
Dissolved copper	94%	Soluble reactive phosphorus	13%	Total vanadium[a]	100%
Dissolved iron	100%	TKN	94%	Total zinc[a]	100%
Dissolved lead	17%	TOC	100%	TSS	96%
Dissolved vanadium	100%	TPH	34%	Turbidity	100%

a Total metal is actually total recoverable

The simulation analysis using SIMPTM focused on these pollutants:

- Total Suspended Solids (TSS)
- Total Chromium (TCr)
- Total Copper (TCu)
- Total Lead (TPb)
- Total Zinc (TZn)
- Total Nitrogen (TN)
- Total Phosphorus (TP)

TSS was selected because it is the primary pollutant simulated by SIMPTM. Other pollutants are related to this simulation. The four metals in the above list were selected based on the review of the stormwater data collected during Phase II. This review suggested that exceedances of either the drinking water or draft receiving water standards may occur at a traffic volume of 100,000 vehicles per day. However, the monitoring data indicated that at the low traffic volume during the monitoring period, of about 10,000 vehicles, only the chromium standard was exceeded. Both the chromium and lead concentrations exceeded the draft Kishon River Standards. Nitrogen and phosphorus were included as they are nutrients.

Considering that the concern regarding stormwater from the Rosh Ha'ayin segment is related to the potential contamination of the drinking water supply of the Hayarkon Springs, it is most appropriate that the Accompanying Team focus on the Israel Drinking Water Standards. However, the draft Kishon River receiving water standards have also been proposed although they may not be valid for either the Einat or Rosh Ha'ayin Channels that receive stormwater from the CIH. Nonetheless, the analysis presented in this report considers both sets of standards.

6.0 MODELING HIGHWAY STORMWATER POLLUTION

Modeling provides an opportunity to simulate the site-specific pollutant accumulation and washoff processes that occur over time, including the removal of accumulated pollutants by various cleaning practices. With modeling a relatively limited set of monitoring data can be used to calibrate the model. The calibrated model is then used to simulate the accumulation and washoff behaviors over a much longer time frame and for conditions that do not currently exist, in particular future traffic volumes.

Highway stormwater pollution results from the accumulation and washoff of contaminated material on pavement surfaces. The sources of these road dirt accumulations are varied and research has shown that their magnitude is directly correlated with the traffic volume the highway experiences. A USEPA study entitled "Contributions of Urban Roadway Usage to Water Pollution" (Shaheen, 1975) examined the variables that affect the deposition of highway surface contaminants by collecting accumulated sediments from nine sites located throughout the Washington, D.C. and Baltimore, Maryland areas. The Average Daily Traffic (ADT) of the monitored sites ranged from zero (a new unopened highway) to 109,000 vehicles per day. The study found strong relationships between the accumulation of highway surface contaminants and the observed traffic volume for the following pollutants: total solids, volatile solids, biochemical and chemical oxygen demand (BOD and COD, respectively), grease, total phosphate, nitrates, nitrites, Kjeldahl nitrogen, chloride, petroleum, asbestos, rubber, lead, chromium, copper, nickel and zinc. For further information on the Shaheen study see Appendix A.

The ability of stormwater runoff to transport sediment and sediment-borne pollutants depends upon many factors. These include amount of the accumulated material and its particle size distribution, the pollutant chemistry or road dirt quality, the intensity and duration of storms, and the physical characteristics of the pavement and roadway such as curbing and landscaping.

The simulation model must have the capability to create a continuous simulation that spans several decades and estimate the storm-by-storm runoff volumes, pollutant concentrations and mass loadings for the various pollutants of interest that would have been discharged from the CIH under an ultimate traffic volume of approximately 100,000 vehicles per day. This long term simulation creates a large population of storm pollutant concentrations that can be analyzed to determine what percentage of those concentrations for a critical pollutant, like total chromium for example, actually exceeded a given water quality standard. The model must also be able to evaluate the pollutant reduction benefits associated with various types and frequencies of highway cleaning practices. The pollution reduction benefits should be measured as a mass reduction in annual pollutant washoff or a decrease in the percentage of storms where a given water quality standard was exceeded.

As stated previously, the SIMplified Particulate Transport Model (SIMPTM) was selected for use on this project. SIMPTM (Sutherland and Jelen, 1998) can simulate the accumulation, washoff and removal processes that occur on streets and highways. It also has the ability to simulate the pollution reduction benefits associated with various cleaning operations. An overview of the SIMPTM model is provided in Appendix B. The benefit of SIMPTM is that it explicitly models the accumulation and washoff of pollutants, unlike other models like the USEPA SWMM model that uses simplified exponential functions.

Most modeling techniques simplistically determine the pollutant loading by multiplying the rainfall runoff of each event times an assumed average pollutant concentration, invariable from storm to storm.

This approach does not provide satisfactory estimates of storm-by-storm concentrations. Additionally, the approach tends to significantly overstate the total pollutant loading In contrast, the SIMPTM model explicitly simulates the physical proceses of stormwater runoff to transport accumulated road dirt and attached pollutants for each storm. The model thereby provides the most accurate simulation of the amount of material washed off during each storm, with an accounting of what remains on the pavement for the next storm. This computation is repeated every six minutes throughout each historic rainfall event. A running summation of these incremental pollutant mass washoffs from each timestep provides the estimated pollutant loading for the storm event.

The overall goals for the modeling of stormwater quality from the Rosh Ha'ayin segment of the CIH are to determine (1) whether water quality standards are being exceeded once the highway's ultimate traffic is realized, (2) how frequently these exceedances may occur, and (3) how various cleaning practices implemented over time on the highway can reduce the frequency of the exceedances to some specified level.

6.1 Modeling Tasks

The key tasks in preparing and using the model were:

1. Obtain and analyze the historic rainfall record from a nearby rainfall gage to establish a simulation time period. The results of this analysis are presented in Section 6.2.
2. Analyze the dry weather road dirt data collected during the summer of 2003 to obtain the accumulation curves for both low and high traffic volumes on both traditional and porous asphalt pavements. Low traffic volumes were those that occurred on the CIH during the May-July 2003 monitoring period. High traffic volumes were those that occurred on Highways 1 and 4 during the same period. The results of this analysis are presented in Appendix C.
3. Analyze the road dirt data to establish the particle size distributions and the chemical qualities of the pollutants of interest to be used in the model calibration in Task 4. The results of this analysis are presented in Appendix D.
4. Using the accumulation curves for low traffic volumes from Task 2 and recent rainfall data from the nearby gage, calibrate the model to best represent the CIH wet weather road dirt accumulation and pollutant concentration data that was observed on both traditional and porous asphalt pavements from October 2002 through April 2003, the wet weather monitoring period.
5. Using the historic rainfall record from Task 1 and the calibrated model from Task 4, simulate the historic long-term storm-by-storm runoff volumes, pollutant concentrations and mass loadings for low traffic volumes.
6. Change the accumulation curves to those that reflect high traffic volumes from Task 2 and re-run the calibrated model from Task 4.
7. Analyze the simulated populations of pollutant concentrations and mass loadings from both Task 5 and Task 6 to obtain summary statistics and identify the frequency of water quality standard exceedances.
8. Use the model from Task 6 to evaluate the pollutant reduction benefits associated with various types and frequencies of street cleaning.
9. Use the model to estimate the pollutant mass load reductions associated with the alternative cleaning practices on both traditional and porous asphalt pavements under both low and high volume traffic conditions.

6.2 Analysis of the Historic Rainfall Record

The SIMPTM modeling package includes a separate program called RAINEV. It is a rainfall analyzer used to evaluate the characteristics of a historic rainfall record. The program divides hourly precipitation records into significant rainfall events and calculates the starting date, duration, depth, time since the last event ended, average intensity, and maximum hourly intensity for all recorded events. The precipitation analysis is a required input file used in the SIMPTM water quality modeling.

For the CIH Stormwater Quality Study, a long-term hourly rainfall record from a station located at Kibbutz Einat immediately east of The Rosh Ha'ayin segment was used. This station provided a 57-year period of rainfall from January 1, 1938 through December 31, 1994 for the model simulations. A 57-year record of rainfall was used for the following reason. For any one future year its particular sequence of storms and storm depths is unknown. Using the historic rainfall record, it is assumed that any future year will mimic a previous historic year of record: that is the future will occur as the past, but the particular year in which this occurs is random and unknown. By simulating loadings based on the historic rainfall record the potential variation in loadings and concentrations is defined to the extent possible. The validity of this approach is recognized when considering that the rainfall during the monitoring period of October 2002 through April 2003 was abnormally and unusually wet. The rainfall was approximately 126% of normal. If only this one time period had been used for simulation, the loadings and concentrations would have been overstated. An analysis of the rainfall during the monitoring period is presented in Section 7.2. The monthly averages found at this location along with the average event for each month is presented in Table 3.

The average number of events is based on all rainfall events recorded with at least 1 mm of rainfall in one hour or 1.7 mm in three hours or 2.5 mm in six hours. The ending of a rainfall event was defined as a period of three hours where no rainfall was recorded.

Data in Table 3 shows that no rainfall occurs during the months of June, July and August and very little has been recorded during the months of May and September. The wet weather season is from October through April, where 99.5% of the annual rainfall occurs.

Table 3 Average Rainfall Characteristics at Kibbutz Einat

	Monthly Averages			Event Averages		
Month	**Number of Events**	**Duration (hrs)**	**Depth (mm)**	**Depth (mm)**	**Duration (hrs)**	**Intensity (mm/hr)**
January	8.5	31.6	78.65	9.23	3.1	2.11
February	7.7	24.8	56.54	7.31	2.8	1.89
March	5.8	15.8	33.37	5.78	2.2	1.71
April	1.5	3.8	9.25	6.28	0.9	1.09
May	0.2	0.5	1.19	5.65	0.2	0.31
June	0.0	0.0	0.00	0.00	0.0	0.00
July	0.0	0.0	0.00	0.00	0.0	0.00
August	0.0	0.0	0.00	0.00	0.0	0.00
September	<.01	0.1	0.28	7.95	0.0	0.09
October	1.2	2.6	8.43	7.17	0.7	1.19
November	4.5	14.3	44.81	9.94	2.0	2.63
December	6.1	22.5	65.57	10.80	2.5	2.14
Average Year	35.5	113.1	298.10	8.40	3.3	2.57

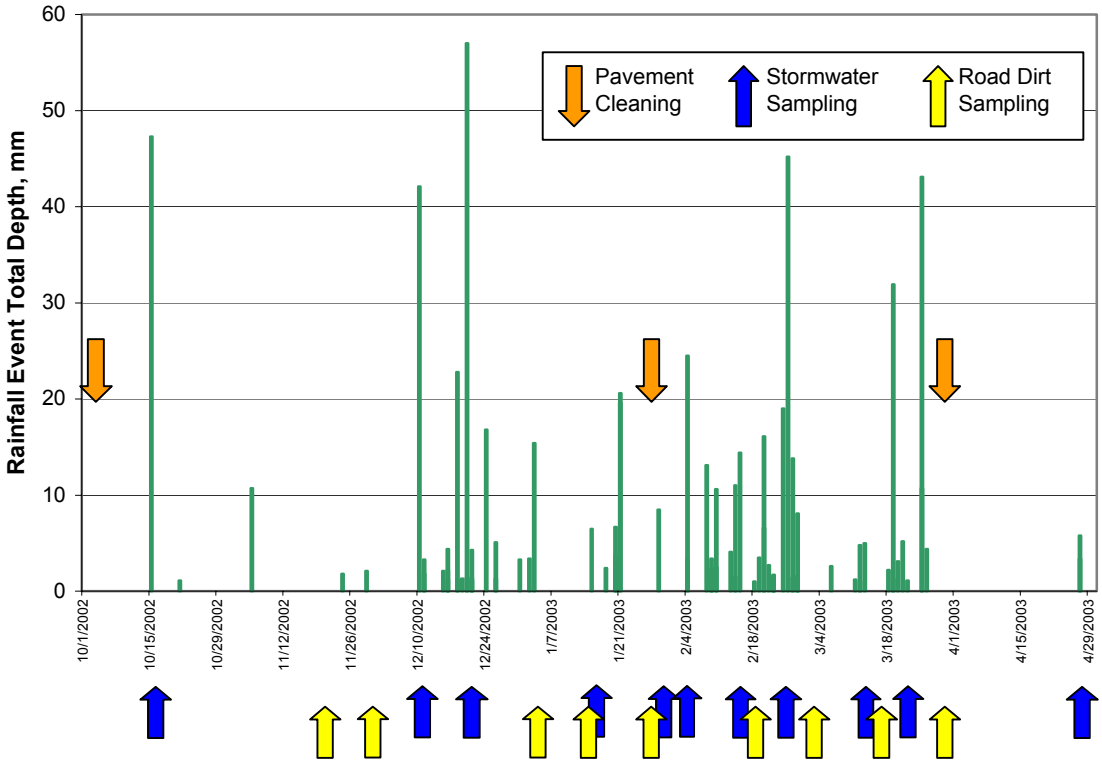
2037 events modeled, 1938 through 1994

7.0 MODEL CALIBRATION

Task 4 of the modeling process is important since proper calibration of the SIMPTM model is key to providing satisfactory simulations. Calibration is derived from the road dirt accumulation and pollutant concentration data observed on both traditional and porous asphalt pavements from October 2002 through April 2003.

The timing of the rainfall events, samplings and cleanings on the CIH during the monitoring period is shown in Figure 2. On October 4, 2002, a two meter wide strip on both the southbound and northbound shoulders immediately adjacent to the New Jersey barrier or curb was first cleaned by the Netivey Hamifratz cleaner (see photo), followed immediately by manual cleaning using an attachment to a vacuum sweeper (see photo) to ensure the gutter next to the barrier was clean. This cleaning procedure was repeated on January 27, 2003 and March 28, 2003 for the northbound shoulder only. During the monitoring period, eleven runoff events were sampled from October 15, 2002 through April 27, 2003. However, the analysis of rainfall data from nearby Kibbutz Einat shows that approximately 78 separate events occurred during this period in which it rained on 51 days. The depths of those events are shown in Figure 2.

Figure 2 Timing of Rainfall Events, Samplings and Cleanings on the Cross Israel Highway



As Figure 2 shows, road dirt sampling occurred nine times between November 14, 2002 and March 30, 2003. Both traditional and porous asphalt pavements were sampled. However, the northbound pavements were only sampled five times from January 13, 2003 to March 16, 2003 whereas the southbound pavements were sampled all nine times. As described in Section 9.1 of the Phase II Monitoring Report, road dirt accumulations along the northbound pavements were heavier throughout the fall and early winter of 2002 due to heavily loaded untarped trucks emanating from a rock quarry in Rosh Ha'ayin spilling sediments along that shoulder. This additional sediment source terminated in January 2003 when CIHC began charging a toll for the use of the highway. As a result, the trucks discontinued their use of the CIH. Because of this additional accumulation on the northbound shoulder, the model calibration focused on the data collected along the southbound shoulder only.

7.1 Model Calibration Steps

The calibration itself involved three steps, each briefly described as follows:

1. Working with the two accumulation curves presented in Appendix C for low volume traffic and with the rainfall during the monitoring period, the accumulation parameters of the model are adjusted to best reproduce the CIH wet weather road dirt accumulations observed on both the southbound traditional and porous asphalt pavements.

2. In an iterative process that uses the results from calibration Step 1, the washoff parameters of the model are adjusted to best reproduce the total suspended solids concentrations observed in the discharges from both the southbound traditional and porous asphalt pavements.

3. Using the final calibrated model for TSS the average pollutant road dirt qualities for the fine, medium and coarse fraction of the accumulated road dirt was adjusted to best reproduce the variation found in the paired concentrations between the pollutant of interest and TSS observed in the discharges from both the southbound traditional and porous asphalt pavements.

7.2 Rainfall Characteristics During the Monitoring Period

Rainfall depth data from the Kibbutz Einat gage was obtained for October 2002 through April 2003. Analysis of the data shows that the monitoring period was very wet when compared to the long-term average rainfall characteristics represented previously in Table 3. Approximately 78 events totaling some 670 mm occurred during the monitoring period, which is 126% more depth than the long term average of 297 mm occurring in an average of 35.3 events. Table 4 shows a monthly comparison of the rainfall recorded during the monitoring period to the long-term averages.

Table 4 Comparison of Rainfall Characteristics Recorded During Monitoring Period to the Long Term Averages

	Number of Events			Rainfall Depth		
Month	Monitoring Period	Historic Average	% Difference	Monitoring Period (mm)	Histroic Average (mm)	% Difference
October	2	1.2	+ 67%	48.4	8.4	+ 476%
November	3	4.5	+ 50%	14.6	44.8	- 67%
December	19	6.1	+ 211%	174.1	65.6	+ 165%
January	10	8.5	+ 18%	88.6	78.7	+ 13%
February	28	7.7	+ 264%	218.6	56.5	+ 287%
March	14	5.8	+ 141%	117.6	33.4	+ 244%
April	2	1.5	+ 33%	9.2	9.3	- 1%
Totals	78	35.3	+ 121%	671.1	296.7	+ 126%

The amount and intensity of rainfall that fell in the months of October, December, February and March was extremely high when compared to the long-term average. As a result, it is quite possible that the pollutant concentrations and/or loadings observed during several of the sampled events during these months were higher than expected during an average year. The results of the model calibration to the collected dataset are presented in Appendix E. This appendix presents graphs comparing observed and simulated concentrations for several pollutants.

7.3 Dissolved Metals Adjustment

As stated previously, SIMPTM simulates concentrations and loadings of the TSS. SIMPTM also simulates the particulate concentration and loadings of other pollutants such as metals based on the observed concentrations of those pollutants in the road dirt. However, SIMPTM does not directly simulate the dissolved fraction of each pollutant. SIMPTM is essentially a sediment particulate washoff model whose ability to simulate other pollutants is a function of the simulation of TSS washoff. Judgment is used, based on the stormwater quality data collected in Phase II, to add an appropriate value for the dissolved concentration for each pollutant of interest. This value is either a constant irrespective of the TSS concentration, or a varied concentration related to the TSS concentrations observed in the monitoring data

During the model calibration process it was determined that the model satisfactorily simulates the particulate fraction of the various metals. However, the model underestimates the metals concentrations that were being observed at lower TSS concentrations. This discrepancy occurs because the fraction of the dissolved pollutant was greater at low TSS concentrations than at high concentrations. To correct for the discrepancy, an adjustment was made to the particulate-based washoff simulations from SIMPTM. For chromium, zinc, and copper, a single average dissolved metals concentration was added to the particulate concentrations simulated by SIMPTM. A dissolved metals adjustment for lead was not needed. Lead occurs primarily in the particulate state. Only 17% of the samples were able to detect dissolved lead (i.e. see Table 2). As a result, the calibrations for total lead were excellent without the need for an adjustment (see Figures E.13 and E.17 in Appendix E).

Table 5 represents the statistical characteristics of dissolved metals data collected during the monitoring study. For each of the three metals in the table, the mean concentration shown was added to the particulate concentrations simulated by SIMPTM.

Table 5 Statistical Characteristics of Observed Concentrations of Dissolved Metals in CIH Pavement Runoff

Statistic	Dissolved Metal		
	Copper (mg/L)	Chromium (mg/L)	Zinc (mg/L)
Mean ($\bar{x}$)	.015	.017	.071
Median	.014	.011	.046
Standard Deviation (SD)	.004	.015	.065
Coefficient of Variation (COV)[a]	.282	.906	.913
70 Percentile[b]	.016	.019	.072
80 Percentile	.017	.026	.079
90 Percentile	.020	.040	.130

a COV = SD / $\bar{x}$
b Percentile is defined as the stated percentage of the observations in the dataset are less than or equal to the number shown.

8.0 MODEL RESULTS

The calibrated SIMPTM model was used to simulate the runoff volume, pollutant loading, and pollutant concentration for each storm through the 57 years of rainfall data. The simulations were for discharges from both traditional and porous asphalt pavements under both low and high traffic volumes, and with and without alternative cleaning practices. As discussed in Section 6.2, the simulation used the 57-year rainfall record of the Kibbutz Einat gage, from January 1, 1938 through December 31, 1994.

8.1 Simulated Pollutant Loadings and Concentrations

Pollutant loadings are the mass of a pollutant transported by stormwater from a designated area over a specified time period. The units are in kilograms per curb-kilometer (kg/curb-km) for streets and highways. In Tasks 5 through 7 of the modeling process (see Section 6.1), the model was used to simulate over the historic rainfall record the storm-by-storm pollutant mass loadings and concentrations. This was done for both low and high traffic conditions with no shoulder cleaning operations. Low traffic volumes were those that occurred on the CIH during the May-July 2003 monitoring period. High traffic volumes were those that occurred on Highways 1 and 4 during the same period. The results of the road dirt analysis for data collected during these dry weather periods are presented in Appendix C.

Pollutant loadings are very important as they are used to compare the magnitude of pollution originating from a source of interest like the CIH to that of other source areas such as agricultural lands, urban lands and other highways. Pollutant loadings can also be used to quantify the water quality benefits of alternative cleaning practices (which is discussed in Section 8.2) by using the SIMPTM model to estimate the reduction in the annual pollutant loadings, which will in turn reduce the loadings and concentrations of other pollutants like chromium. Although pollutant loadings are important, most scientists concerned with stormwater pollution prefer to sample and evaluate the concentrations of pollutants in stormwater since these values can be easily compared to various water quality standards like those for drinking water.

Table 6 presents the average annual pollutant loadings for discharges from the Rosh Ha'ayin segment's asphalt pavements over the 57-year simulation. The table shows that a ten-fold increase in traffic volume will, depending on the pollutant, results in an average increase of 291% to 315% in annual mass loadings from the CIH.

Table 6 Average Annual Pollutant Loadings[a] and Mean Concentrations for the CIH Under Low[b] and High[c] Traffic Volume Conditions with No Cleaning

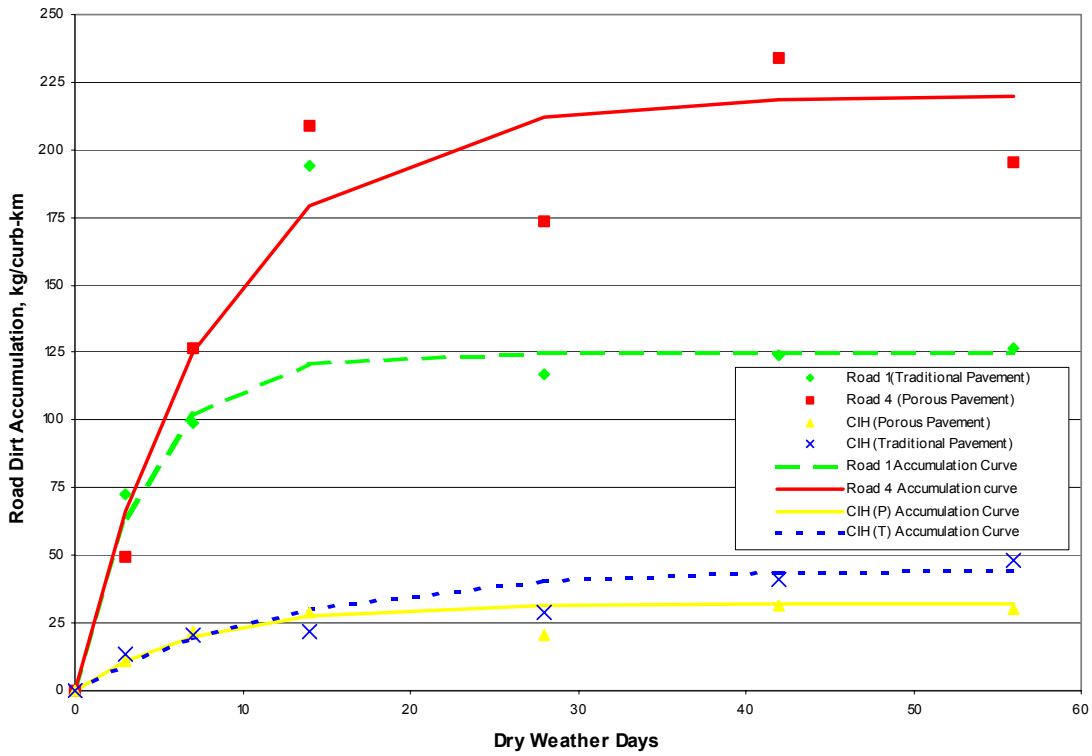
Pollutant	LOADINGS		Percent Increase	CONCENTRATIONS		Drinking Standards Mg/L	Draft River Standards Mg/L
	Low Traffic	High Traffic		Low Traffic	High Traffic		
TSS	498	1976	297	112	443	NS[d]	NS
TCr	.017	.067	294	.021	.032	0.05	0.01
TCu	.012	.048	300	.018	.026	1.40	0.05
TPb	.056	.231	313	.013	.052	0.10	0.01
TZn	.054	.224	315	.083	.121	5.00	1.00
TN	1.63	6.37	291	.365	1.43	NS	NS
TP	.301	1.22	305	.067	.274	NS	NS

a kg/curb-km/yr
b Low Traffic is approximately 10,000 vehicles per day
c High Traffic is approximately 100,000 vehicles per day
d. NS means no standard

Table 6 indicates that the annual pollutant loadings increase by only a factor of approximately three although the volume of traffic increases by a factor of 10. How is this possible? There are two reasons. First, as shown in Figure 3, the amount of road dirt that accumulates on highways is not directly

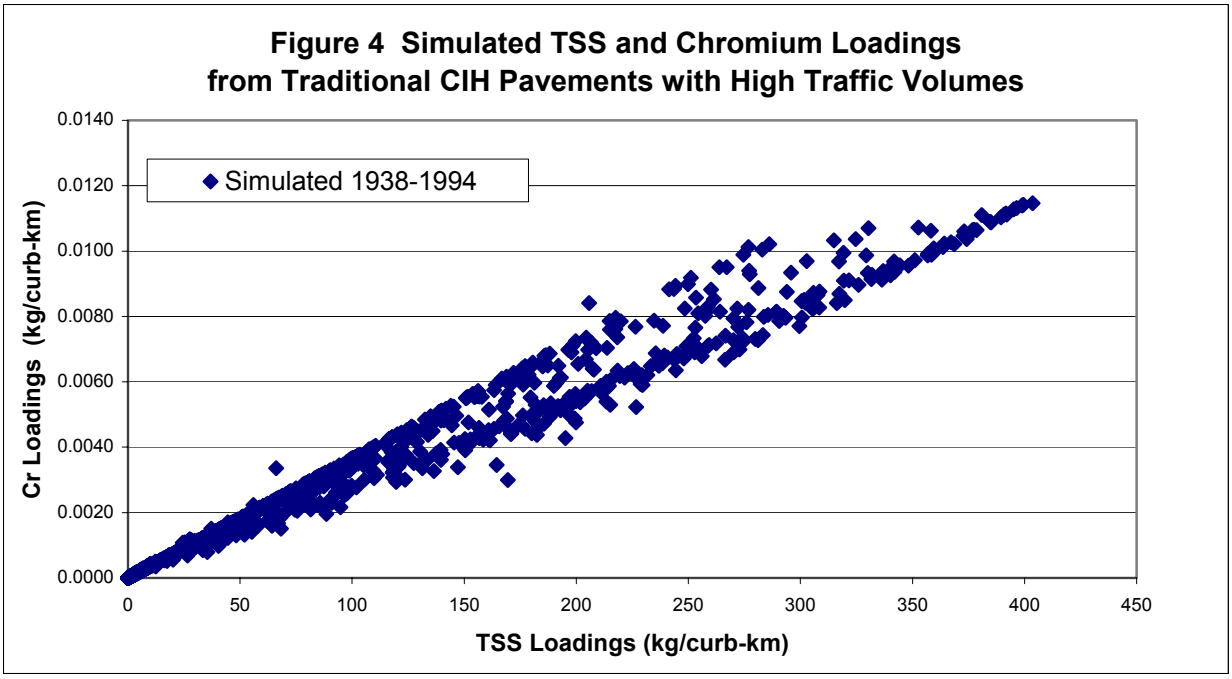
proportional to the traffic volume. Secondly, at higher quantities of accumulated road dirt most of the storms have insufficient volumes to remove all of the accumulated material. The general relationships between traffic volume and road dirt accumulation in Figure 3 are consistent with other studies of highways in the United States and Europe (Drapper, et al., 2000; Driscoll, et al., 1990; Hoffman, et al., 1985; Horner and Mar, 1984; Irish, et al., 1998; Kobriger, N.P., 1981; Shirley, et al., 1982; Thomson, et al., 1997; Wu, et al., 1998).

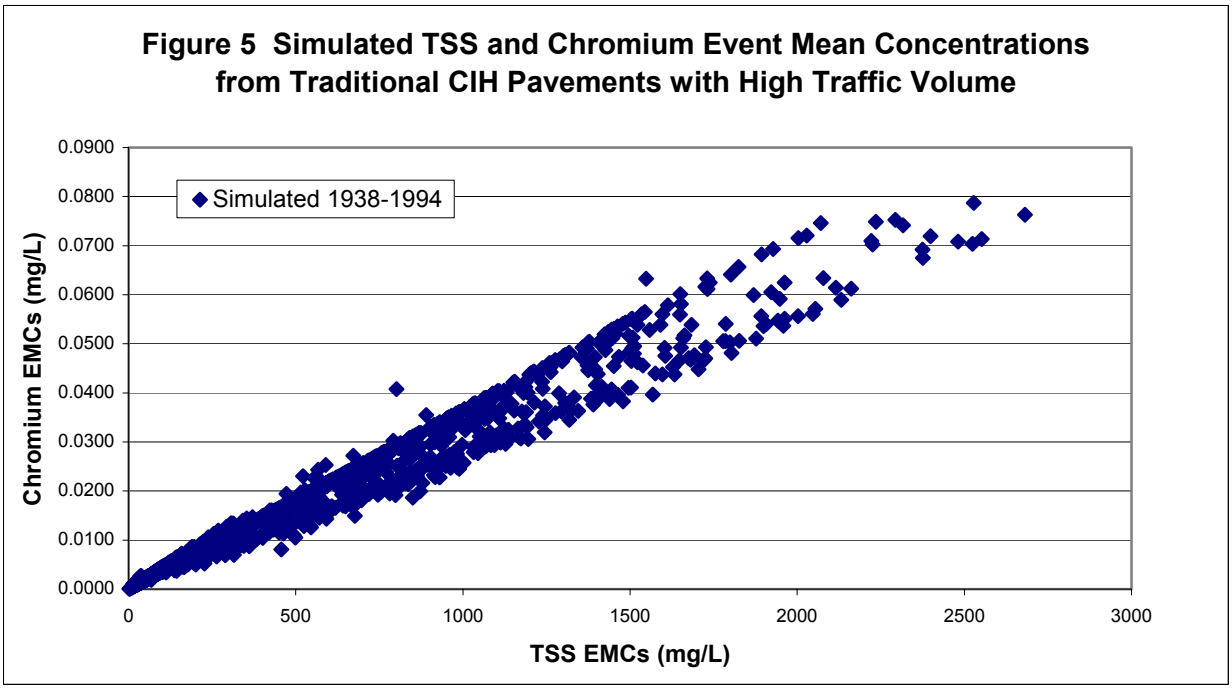
Figure 3 Dry Weather Road Dirt Accumulation



The average annual loadings shown in Table 6 were obtained as follows. The individual mass loadings for each of the 2037 storms washed off of a representative area of the Rosh Ha'ayin segment that drains to a single spillway were summed. This total was divided by both the curb-kilometers of the tributary drainage area and by the 57 years of the simulation period.

Figures 4 and 5 presents the paired TSS and chromium simulations for all of the 2037 storms for loadings and concentrations, respectively. The simulations are for a traffic volume of 100,000 vehicles per day with no pavement cleaning. Figures 4 and 5 show that there is a relationship between TSS loadings and chromium loadings, and in turn, concentrations. Therefore, the strategy to reduce the concentration of chromium is to reduce the TSS loading.





The statistical characteristics of the individual TSS and chromium mass loadings for the 2037 storms shown in Figure 4 are presented in Table 7.

Table 7 Statistical Characteristics of Simulated TSS and Chromium Storm Loadings from Traditional CIH Pavements with High Traffic Volume

Statistic	Simulated Storm-by Storm Loadings	
	Total Suspended Solids (TSS) (kg/curb-km)	Chromium (kg/curb-km)
Mean ($\bar{x}$)	53.7	.00167
Median	7.39	.00028
Standard Deviation (SD)	88.8	.00263
Coefficient of Variation (COV)	1.65	1.58
80 Percentile	97.5	.00032
90 Percentile	196.5	.00598
95 Percentile	266.3	.00797

The statistics in Table 7 indicate that the average TSS loading for all 2037 storms is 53.7 kg/curb-km. However, the median loading, the loading that would be ranked 1019 out of 2037 storms is only 7.39 kg/curb-km. When viewing the actual distribution of these individual storm loadings shown in Figure 4, keep in mind that several hundred of these loadings have been plotted on top of each other. In fact, 80% of the storm loadings, or approximately 1600 points, have been plotted at or less than a TSS loading of 97.5 kg/curb-km and over 1000 points are plotted at or below 7.4 kg/curb-km. The distribution of these storm loadings start to thin out at the upper end of the plot. Although the points still appear to be densely distributed in this upper area, only 10% of the loads (i.e. 204 out of 2037) are plotted at a TSS loading greater than 196.5 kg/curb-km.

The statistical characteristics of the individual TSS and chromium concentrations for the 2037 storms are presented in Table 8. The chromium concentrations have been adjusted by adding a constant dissolved chromium concentration of 0.017 mg/L to each EMC as discussed in Section 7.3.

Table 8 Statistical Characteristics of Paired TSS and Chromium Event Mean Concentrations from Traditional CIH Pavements with High Traffic Volume

Statistic	Simulated Event Mean Concentrations	
	Total Suspended Solids (TSS) (mg/L)	Chromium (mg/L)
Mean ($\bar{x}$)	430	0.031
Median	238	0.026
Standard Deviation (SD)	479	0.015
Coefficient of Variation (COV)	1.11	1.05
80 Percentile	776	0.042
90 Percentile	1120	0.054
95 Percentile	1451	0.064

The statistics in Table 8 suggest that chromium concentrations in stormwater from the traditional pavements of the Rosh Ha'ayin segment under the projected conditions of approximately 100,000 vehicles per day will exceed the Israeli drinking water standard of 0.05 mg/L for approximately 13.3% of the storms or 271 out of 2037 storms that occurred during the 57-year simulation period. This represents an average of about five exceedances per year. In other words, the Israeli drinking water standard for chromium will be complied to for 86.7% of the storms under the conditions of 100,000 vehicles per day with no pavement cleaning. Absent the adjustment of the dissolved fraction as previously described, only 4.1% or 84 out of the 2037 concentrations simulated by the model exceed the standard.

Regarding lead, copper and zinc: the simulations indicate that the frequency of standard exceedances for lead are similar to chromium, on the order of 13%. However, this simulation does not recognize the long-term effects of the use of unleaded gas, which will diminish the concentrations of lead in stormwater. Hence, the simulations overstate the likely frequency of future exceedances. In contrast, the concentrations of copper and zinc did not exceed the drinking water standard in any of the 2037 simulated storms.

With respect to the proposed Kishon River receiving water standard, exceedances for chromium would occur for 100% of the storms at 100,000 vehicles per day. Absent the adjustment to the dissolved fraction previously described, approximately 46% or 937 of the 2037 chromium concentrations simulated by the model exceed the draft receiving water standard. This represents an average of 16.4 exceedances each year.

8.2 Load Reductions From Cleaning Practices

In Tasks 8 and 9 of the modeling process (see Section 6.1) the simulations over the historic 57-year rainfall record were of pollutant load reductions achieved from pavement cleaning practices. The ability of a street sweeper or cleaner to reduce overall pollutant washoff loads depends on several factors. First and foremost is the cleaner's innate ability to remove accumulated road dirt. Another factor is the complex environmental dynamics of road dirt accumulation between storm events and the washoff during each storm event. As shown in Appendix E, the SIMPTM model can accurately simulate this complicated interaction of road dirt accumulation and washoff. Consequently, the interaction of street cleaning throughout the simulated period is a logical extension of the model's capabilities. Appendix F documents the street cleaning component of the SIMPTM model and the parameter values used to simulate the pollutant reductions on the CIH for the cleaning technologies that were evaluated. Four street cleaning technologies were evaluated for use on the CIH, specifically newer mechanical sweepers, regenerative air (vacuum) sweepers, the Schwarze EV sweeper, and the Netivey Hamifratz pavement cleaner. Table 9 documents the specific cleaning dates assumed in Tasks 8 and 9 of the modeling process. All four technologies are described in detail in Appendix F.

The analysis presented in Appendix F shows that the most effective technology is the Netivey Hamifratz pavement cleaner followed closely by the Schwarze EV and the regenerative air or vacuum sweeper. The least effective cleaner is the mechanical sweeper. However, the most cost effective cleaner is likely the regenerative air or vacuum sweeper. If it is decided to clean the Rosh Ha'ayin segment of the CIH, the selected cleaning equipment, if other than the Netivey Hamifratz (NH) pavement cleaner, should be tested to verify its pick-up performance is equivalent or better than that assumed (see Appendix F for details). In addition, hand cleaning of the gutters along the sections of porous pavement should be conducted in the same manner that occurred when the highway was cleaned during the monitoring period.

Table 9 Cleaning Dates Assumed for SIMPTM

Number of Cleanings per Wet Season	Cleaning Dates each Year
3	October 15th
	December 15th
	February 15th
4	October 15th
	December 1st
	January 15th
	March 1st
5	October 15th
	December 1st
	January 1st
	February 1st
	March 1st
6	October 15th
	November 15th
	December 15th
	January 15th
	February 15th
	March 15th

8.3 Impact of Cleaning Practices on Chromium Concentrations

As discussed in Section 8.1, it was estimated that approximately 13% of the storms under a projected 100,000 vehicles per day may exceed the Israeli drinking water standard for chromium of 0.05 mg/L. If the objective of a cleaning program on the CIH was to reduce the chromium loadings so that the Rosh Ha'ayin segment pavement discharges will not exceed the Israeli drinking water standard more than 5% to 10% of the time, for example, the needed reduction in the TSS loads could be on the order of 10% to 30%. A detailed examination focused on the impact these four cleaning practices have on the distribution of chromium concentrations in the CIH pavement discharges. The specific cleaning dates assumed for the SIMPTM simulations were shown previously in Table 9. Using these cleaning dates, the model simulated the 2037 chromium concentrations that would occur assuming the sixteen alternative cleaning operations (i.e. four cleaning technologies times four different cleaning frequencies) were implemented throughout the 57-year simulation period. Table 10 presents a summary of the effect of these sweeping practices on the frequency of chromium exceedances of the Israeli Drinking Water Standards.

Table 10 Effect of Sweeping Practices on the Frequency of Compliance with the Israeli Drinking Water Standards

TECHNOLOGY	SWEEPING FREQUENCY				
	0	3	4	5	6
Mechanical	86.7%	90.6%	92.0%	93.0%	93.8%
Regenerative air	86.7%	90.6%	92.2%	93.3%	95.0%
Schwarze	86.7%	90.6%	92.0%	93.3%	94.4%
NH	86.7%	91.1%	92.8%	93.9%	95.7%

Loading reductions were estimated from SIMPTM for cleaning practices that resulted in alternative frequencies of compliance with the drinking standards. Tables 11 and 12 present these reductions for 90% and 95%, respectively. Table 11 reflects the use of a regenerative air or vacuum sweeper to clean the Rosh Ha'ayin segment three times each wet weather season on the dates shown in Table 9. Table 12 reflects the use of the same type of sweeper six times each wet weather season on the dates shown in Table 9.

Table 11 Average Annual Pollutant Loadings[a] for CIH under Low[b] and High[c] Traffic Volume Conditions with 90% Compliance

Pollutant	Traditional Pavements				Porous Pavements			
	Low Traffic	% Reduction[d]	High Traffic	% Reduction[d]	Low Traffic	% Reduction[d]	High Traffic	% Reduction[d]
TSS	411	13	1695	12	481	10	1740	16
TCr	.0123	13	.0513	14	.0182	11	.0623	20
TCu	.0092	12	.0381	13	.0131	11	.0429	19
TPb	.0414	13	.173	14	.060	11	.225	19
TZn	.0416	12	.172	13	.0585	10	.217	18
TN	1.39	13	5.72	11	1.51	10	5.39	14
TP	.236	13	.978	13	.309	11	1.14	17

a kg/curb-km/yr
b Low Traffic is approximately 10,000 vehicles per day
c High Traffic is approximately 100,000 vehicles per day
d Reduction from not cleaned loadings

Table 12 Average Annual Pollutant Loadings[a] for CIH under Low[b] and High[c] Traffic Volume Conditions with 95% Compliance

Pollutant	Traditional Pavements				Porous Pavements			
	Low Traffic	% Reduction[d]	High Traffic	% Reduction[d]	Low Traffic	% Reduction[d]	High Traffic	% Reduction[d]
TSS	295	38	1291	33	369	31	1300	37
TCr	.0091	35	.0413	31	.0144	30	.0485	37
TCu	.0068	35	.0307	30	.0104	29	.0332	37
TPb	.0308	35	.139	31	.0477	29	.180	35
TZn	.0310	34	.139	29	.0466	28	.176	34
TN	.984	39	4.27	34	1.14	32	3.89	38
TP	.172	37	.765	32	.242	30	.885	36

a kg/curb-km/yr
b Low Traffic is approximately 10,000 vehicles per day
c High Traffic is approximately 100,000 vehicles per day
d Reduction from not cleaned loadings

9.0 DISCUSSION

Two primary questions of key importance should be addressed.

Question 1: Will reducing the frequency of the water quality standards exceedances significantly improve the quality of water in the Einat and Rosh Ha'ayin Channels?

Examination of the data collected during Phase II helps to address this question. Stormwater sample were taken from the two channels at stations upstream of The Rosh Ha'ayin segment. The data indicate that with respect to chromium the Israeli Drinking Water Standards were exceeded in four of 16 samples, or 25%, and the draft Kishon Standards were exceeded in 13 of 16 samples, or 81%. As previously stated, the model simulations indicate the chromium standard may be exceeded 13% and 100% for the drinking water and draft river water standards respectively.

The Rosh Ha'ayin segment of the CIH pavement represents only 1% of the total watershed area that drains to Einat and Rosh Ha'ayin Channels. Other uncontrolled sources of pollution exist in the watershed that are much greater in area and pollutant loading, such as agriculture, highways, and commercial and residential areas. Failure to control these other sources would negate any benefits associated with reducing the pollutants from the CIH.

Therefore, it is concluded that reducing the frequency of exceedances from The Rosh Ha'ayin segment will not likely improve water quality in the Einat and Rosh Ha'ayin Channels. Furthermore, in the absence of any national Israeli policy on the control of pollutants in stormwater from all nonpoint sources, it would be a waste of public money to treat stormwater from the CIH while not addressing the other more dominant discharges.

Question 2: If we wish to increase the compliance frequency for the water quality standards, what highway cleaning practices would be needed?

For the Israeli Drinking Water Standards with respect to chromium, three cleanings each year with a regenerative air or a modern vacuum street sweeper would be needed to increase compliance from 87%, that is already achieved without sweeping, to 90%. Increasing compliance to 95% would require that the pavement be swept six times per year.

For the draft Kishon River Standards, highway cleaning practices alone would not be able to increase the compliance frequency. This is because cleaning practices are only marginally effective at removing dissolved pollutants. The monitoring data indicate that the mean and median concentrations of dissolved chromium are 0.017 and 0.011 mg/L, respectively, under low traffic volume. These values exceed the proposed standard of 0.01 mg/L. To meet the standards at a reasonable frequency of compliance would therefore require both extensive sweeping and a stormwater treatment system that is capable of removing dissolved metals at considerable expense.

10.0 CONCLUSIONS

The review of the monitoring data and model simulation with SIMPTM leads to the following conclusions.

1. The quality of stormwater, the accumulation of road dirt and its chemistry of the Rosh Ha'ayin segment is typical of stormwater quality observed from freeways in other countries with similar traffic volumes.

2. It is estimated that without pavement cleaning, the quality of the stormwater will over the long term comply with the Israeli Drinking Water Standards for 87% of all storms, but 0% with respect to the draft Kishon River standards.

3. Currently, the observed quality of water in the Einat and Rosh Ha'ayin channels above the CIH fails to meet either the drinking water or draft river standards at similar frequencies to the stormwater from the CIH.

4. Compliance of the stormwater from the CIH with the drinking water standards can be increased from 87% to 90% by cleaning the Rosh Ha'ayin segment three times per year during the wet season; 95% if cleaned six times per wet season each year.

5. Because the Rosh Ha'ayin segment pavement represents only 1% of the watershed area draining to the two channels, cleaning the CIH within the watershed will not provide noticeable water quality benefits.

6. The containment basins and other practices by CIHC and Derech Eretz, Ltd (the operator) provides satisfactory prevention of the potential harmful effects of hazardous chemical spills from accidents on the CIH.

11.0 RECOMMENDATIONS

1. Absent a national or watershed policy that leads to the reduction of non-point pollution from all sources in the watershed of the Einat and Rosh Ha'ayin channels, it is recommended that the Rosh Ha'ayin segment of the CIH not be cleaned or swept at this time.

2. If it is decided to clean the Rosh Ha'ayin section of the CIH, the selected cleaning equipment, if other than the Netivey Hamifratz (NH) pavement cleaner, should be tested to verify its performance. In addition, hand cleaning of the gutters along the sections of porous pavement should be conducted in the same manner that occurred when the highway was cleaned during the monitoring period.

3. To verify the simulations of SIMPTM, it is recommended that the CIH Ltd conduct further monitoring when the traffic volume reaches approximately 100,000 vehicles per day. The types of data collected should be specific to the needs of verifying the model simulations. The detailed monitoring plan should be prepared when the volume of traffic approaches 100,000 vehicles.

4. Current measures to protect against the potential harmful effects of accidental hazardous chemical spills should be continued.

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